

Exploring the associations between play types and fundamental motor skills: a cross sectional study

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INTRODUCTION

Fundamental motor skills (FMS) are basic movement patterns formed in childhood that shape more advanced movements that are used in sports and physical activity (PA). They include locomotor skills (e.g. running, jumping, skipping, etc) and object control (OC) skills (e.g. throwing, catching, kicking, etc) [1]. Multiple studies have explored the importance of developing FMS in early childhood, with findings suggesting that FMS proficiency is associated with fitness, physical literacy, PA participation and even cognitive and academic performance [2,3,4]. Given the significance of FMS, it is important that young children have the chance to develop their FMS proficiency (and have adequate competency by age 7) [1].

Young girls are more likely to have worse total FMS proficiency and OC skills than boys [4]. This is a concern as lower FMS proficiency and OC skills in childhood are linked with lower moderate intensity PA levels in kindergarten children [3,4] and lower object control skills are linked with lower PA participation in later adolescence [5]. It has been suggested that these FMS gender differences are not due to biological reasons but rather their activities and environments [1,2], and that these differences can be reduced with equal levels of coaching and opportunity [4]

Children exhibit gendered-based differences in their preferred types of play. Boys tend to play with things like toy vehicles, play weapons and manipulative vehicles, while girls tend to gravitate towards dolls, stuffed animals and domestic toys [6].

As FMS in early childhood is more likely to be influenced by social and environmental factors rather than biological, the differences in play preferences could be affecting the FMS of girls - specifically in object handling skills. To date, there has been little research on the link between play preferences and FMS proficiency.

AIMS

The first aim of this study is to examine the associations between children's play and FMS. The second aim is to examine the associations between gender and FMS. The final aim is to examine whether gender influences the relationship between play and FMS.

METHODS

A cross-sectional design will be used to compare FMS proficiency and play preferences of approximately 85 kindergarten children. Demographic data will include age and gender. Their FMS proficiency will be assessed using the Test of Gross Motor Development 3 (TGMD-

3). The TGMD-3 involves testing 13 skills that are divided into 2 categories of locomotor and ball skills. Each skill has 3-5 performance criteria that award 1 point for passing the criteria, and 0 points for failing the criteria. Points are then added up to give locomotor, ball, and total TGMD-3 scores. The locomotor category has 46 possible points while the ball category has 54 possible points. The total range of possible points is 0-100 [7].

Play preferences for each child will be assessed through a play preferences questionnaire completed by each kindergartener's usual teacher. This questionnaire will be modelled after the Preschool Play Preferences Questionnaire used by Lauer et al. [6]. It will involve 12 items total with 6 toys and 6 activities. Half of the items will be "male-typed" and the other half "female-typed". Each item's frequency of play will be scored on a 5 point Likert scale ranging from "never" to "very often". A gender-typed play score will be calculated by adding points from male-typed items and subtracting points from female-typed items. Scores can range from -30 to 30, with a positive score representing preference for male-typical play, and a negative one representing preference for female-typical play.

Multiple linear regression with interaction terms will be used to find the relationship between total TGMD-3 scores and play scores, and whether gender affects that relationship. Object control and locomotor skill scores will also be analysed as subsets of the total TGMD-3 scores.

SIGNIFICANCE

Given there is little evidence on the relationship between play types and FMS, findings from this study will inform and give direction to future research. Finding associations between play types and FMS will lay the groundwork for future longitudinal or RCT studies to help solve the growing issue of motor incompetency and low PA levels of children, especially girls.

REFERENCES

- [1] Bolger L et al. *J Sports Sci* **39**: 717-753, 2020.
- [2] Eather N et al. *Prev Med Rep* **11**: 191-195, 2018.
- [3] Jones et al. *J Sport Health Sci* **9(16)**: 542-552, 2020.
- [4] Zheng Y et al. *Int J Environ Res Public Health* **19(14)**: 8318, 2022.
- [5] Barnett L et al. *J Adolesc Health* **44(3)**: 252-259, 2009.
- [6] Lauer J et al. *Dev Psychol* **54(4)**: 613-620, 2018.
- [7] Webster E & Ulrich D. *J Mot Learn Dev* **5(1)**: 45-58, 2017.